# Project Progress Tracking

## Completed Tasks

- [x] Analyzed the initial `robot_arm_description` URDF and workspace.
- [x] Identified MoveIt 2 and Gazebo Harmonic integration issues related to `<mimic>` joints.
- [x] Outlined strategy to fix mimic joints: use "Passive Joints" in MoveIt 2 and isolate the active driver joint in `ros2_control`.
- [x] Created `updates.md` to document the strategy.
- [x] Created a new ROS 2 workspace package: `robot_arm_gazebo`.
- [x] Created `gazebo.launch.py` to spawn the robot in Gazebo Harmonic.
- [x] Created `rviz.launch.py` for visual joint debugging.
- [x] Created `gazebo_rviz.launch.py` to launch both physics and visualization.
- [x] Added `install` directives to `CMakeLists.txt` for the new directories.
- [x] Replaced legacy `libgazebo_ros_control.so` with `gz_ros2_control-system` in the URDF.
- [x] Solved missing mesh errors by exporting `GZ_SIM_RESOURCE_PATH` and substituting `model://` for `package://` inside the URDF.
- [x] Solved mimic joint capability in Gazebo by passing `--physics-engine gz-physics-bullet-featherstone-plugin` to the launch file.

## Current Issues & Roadblocks

- [x] **Colcon Build Failure:** CMake cannot find `gazebo_ros2_control`. Fixed by switching to Harmonic integrations (`gz_ros2_control`).
- [x] **ABI Mismatch:** `libgz_ros2_control-system.so` undefined symbol. Fixed by updating system ros-jazzy packages via `apt` update.
- [ ] **Ogre Material Warnings:** The legacy material "Gazebo/Silver" generates warnings in Gazebo Harmonic. This isn't fatal as it falls back to default, but eventually, standard modern PBR materials should be applied.

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## Next Steps

- [x] Fix the `<transmission>` tags in `robotic_arm.trans` (currently ROS 1 style) to ROS 2 standard `<ros2_control>` tags.
- [x] Configure the `ros2_control` hardware interface strictly for the active gripper joint (ignoring the passive ones for command).
- [ ] Create controllers (`joint_state_broadcaster`, `joint_trajectory_controller`) in a `controllers.yaml` file.
- [ ] Integrate Controller execution to the launch file.
- [ ] Re-run MoveIt 2 Setup Assistant to generate the updated SRDF using our "Passive joint" strategy.